

Irrigation Automation Opportunities in Rural Areas

The irrigation sector is abuzz with wireless applications. A tremendous market potential is visible and technology solutions are also available



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In the era of Internet, mobiles, GPS, Ethernet and 3G, technology has reached urban, industrial and some rural belts. But as far as rural irrigation is concerned, there is a huge gap to be filled in yet.

There are two types of rural irrigation systems: Drinking water supply schemes (consisting of water pump, tanks, valves, etc) and agriculture water supply systems (consisting of water pump, pipe line network in the farms, valves, etc). Traditional water pumps are controlled by a simple starter with on/off pushbuttons. Nowadays, the trend is to automate irrigation systems using basic electronic devices like single-phasing preventer, timer and remote control for water pumps.

There are several pains in operating the irrigation systems in traditional

ways. Water pumps, tanks and farms are distantly located at different places away from the operator's house. The irrigation operator has to operate these systems against a series of hurdles like erratic power supply, long travel over difficult terrain and fear of animals on the way to pumps.

These hurdles create inefficiency in the pump operations leading to heavy wastage of water and electricity. Excess water erodes the soil and damages the civil structures of water tanks. All in all, there is a huge loss of energy in many ways. Considering this, automating the irrigation sector is vital.

During the last decade, there has been a major innovation with irrigation witnessing the use of cellphones for controlling the water pumps. Since 2004, a farmer could switch on/off water pumps using a mobile phone. With some inertia in the initial period, the market accepted the technology after realising the utility of the product.

After 2009, many players in the automation industry sought opportunities in irrigation applications with multiple features in existing technologies. This gave birth to GSM modem-based irrigation automation solutions, eliminating the use of mobile phones. However, only institutions and rich farmers could afford this advanced technology, while masses are still insisting on low costs. Government institutions are adopting latest technology like supervisory control and data acquisition (SCADA) in water purification plants, but there is least penetration of technology in the rural private sector.

These days, GSM alerts are provided for pump theft, water level and pump failure on the mobile phone. A few companies have developed systems to read the current electrical status of a water pump. There are even systems to control the pumps based on the moisture requirement of the soil, but these are being used in some agriculture institutes and corporate farming only.

In western countries, irrigation is computerised, which is not possible in India until there is assurance of continuous electricity supply. Government bodies have attempted solar pump-based irrigation in some states, but these are not much popular in open market due to their high capital costs. A few multinationals conducted pilot testing of their soft starters and GSM systems for the Indian rural market, but these were reported as failures due to odd conditions in rural irrigation systems.

In short, the irrigation sector is abuzz with wireless applications. A tremendous market potential is visible and technology solutions are also available. Big opportunities are inviting technocrats to rural India. But irrigation market is very sensitive and challenging. Its characteristic is exponential—demand being slow initially but huge over a period of time. In order to reach the technology to masses, it has to be low-cost with assurance of support at local level. There must be commitment and passion to work in this sector and time has come for the techies to move from urban to rural area for working on a drop of water—a vital need of the time! ●